

Predictive Modeling for Early Diagnosis of Alzheimer's Disease Using Machine Learning & Data Mining

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Note: The Final Report is 8 Pages in total. The First and Last page are the Table of Contents and References.
The main content of the report is 6 pages.

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Abstract

Alzheimer's Disease is a complex neurological condition that presents significant challenges in early diagnosis. This research explores the application of machine learning and data analysis techniques to develop predictive models for identifying early Alzheimer's Disease stages (*C Kavitha I, 2022*). By utilizing data from the Open Access Series of Imaging Studies (OASIS1 and OASIS3), the study incorporated MRI-derived features and cognitive assessment scores like the Mini-Mental State Examination (MMSE) to enhance diagnostic understanding and accuracy (*Neurosci, 2010*). Multiple machine learning models were strategically deployed to predict dementia stages, including Logistic Regression, Support Vector Machine, Random Forest, and XGBoost. To augment model performance, ensemble techniques such as Voting Classifiers were implemented. The analysis revealed that longitudinal data from the OASIS3 dataset substantially improved prediction accuracy, particularly in detecting severe dementia stages. Nevertheless, challenges persist, including imbalanced datasets and difficulties in predicting less common dementia stages, indicating the need for continued methodological refinement.

Explainable AI techniques, specifically SHapley Additive exPlanations (SHAP), were integrated to provide transparent insights into feature contributions. By highlighting the significance of brain volume and cognitive assessment scores, these explanations offer clinicians deeper understanding of risk factors and disease progression mechanisms (*Khandaker Mohammad Mohi Uddin I, 2023*).

The research demonstrates machine learning's transformative potential in advancing Alzheimer's diagnosis and care. Future research trajectories will focus on implementing advanced computational techniques like deep learning, addressing data imbalance challenges, and ensuring fair, accurate outcomes across diverse patient populations.

Keywords

Alzheimer’s disease, machine learning, predictive modeling, early diagnosis, dementia stages, data mining, OASIS dataset, MRI features, Mini-Mental State Examination, longitudinal data, feature selection, class imbalance, random forest, XGBoost, explainable AI, SHAP, cognitive assessment, ensemble methods, neuroimaging, diagnostic accuracy (*Hala Alshamlan *ORCID, 2024*).

Introduction

Alzheimer Disease is a condition that gradually takes away the memory of a person, ability to think, leaving an impact on those who are suffering from it and people who take care. This is why identifying the disease at an early stage is very important. The families can be prepared for the challenges (*Aristidis G Vrahatis I, 2023*). The methods like clinical assessments and brain scans aren’t proving to be enough. They are expensive and don’t provide accurate results. This is where technology enters the picture.

Advancements in techniques using machine learning and data mining are proving to be vital. They can gather data that might not be obvious to the human eye. By using insights from these, we can create models that can predict early stages of dementia and help doctors take better decisions (*Vijeeta Patil, 2022*).

In this project, we worked on data from Open Access Series of Imaging Studies that includes information of people at various stages of cognitive health (*Shaofan Jiang, Shaofan Jiang1, & Siyu Yang3, 2024*). The data has brain imaging results,

scores from tests like the Mini-Mental State Examination (MMSE), age, gender etc. We developed machine learning models using this rich dataset that predicts early stages of Alzheimer's and highlights key factors of the predictions.

We have incorporated AI (XAI) techniques like SHapley Additive exPlanations (SHAP). These methods help doctors understand why the models make certain predictions, making the technology more transparent.

To conclude, we would like to make diagnosing Alzheimer's easier, earlier, and more accurate. By blending advanced technology with medical needs, we hope to ease the burden of this disease and improve the lives of patients. It's about creating tools that don't just work but truly make a difference (*Ling Yue # 1, 2023*).

Methods

A structured methodology was adopted to design predictive models for Alzheimer's Disease (AD) diagnosis, utilizing machine learning and data analysis techniques. The key steps in this process are outlined below.

Data Collection

The datasets used in this study were obtained from the Open Access Series of Imaging Studies (OASIS), comprising:

a) OASIS1 (Cross-Sectional): Contains demographic information, cognitive assessments such as the Mini-Mental State Examination (MMSE) and Clinical Dementia Rating (CDR), along with MRI-based features like estimated Total Intracranial Volume (eTIV), normalized Whole Brain Volume (nWBV), and Atlas Scaling Factor (ASF).

b) OASIS3 (Longitudinal): Tracks participants over time, providing data on changes in cognitive function and brain imaging metrics.

Both datasets underwent careful preprocessing to maintain consistency and reliability.

Data Preprocessing

Steps taken to prepare the data include:

a) Cleaning: Incomplete, missing, or redundant records were excluded to ensure data quality.

b) Feature Encoding: Categorical variables were converted into numerical representations to enable machine learning analysis.

c) Normalization: Continuous variables were standardized to ensure a uniform scale across features.

d) Data Splitting: The data was divided into training (80%) and testing (20%) subsets for model building and validation.

For the OASIS3 dataset, additional temporal variables were included to account for longitudinal trends, which are important for studying disease progression.

Feature Selection

Feature selection methods were applied to identify variables with the highest predictive value. This step improves the efficiency and interpretability of the models. Techniques used include:

a) Correlation Analysis: Identified relationships between features and the target variable.

b) Information Gain: Measured the importance of each feature for predicting outcomes.

c) Chi-Square Tests: Evaluated statistical independence between features and target variables.

Significant predictors included MMSE scores, nWBV, age, and MRI-based features.

Model Development

Several machine learning algorithms were explored to create robust predictive models:

a) Logistic Regression: Provided a baseline for comparison.

b) Support Vector Machines (SVM): Modeled complex relationships in high-dimensional data.

c) Random Forest: Managed large datasets effectively and highlighted feature importance.

d) XGBoost: Addressed class imbalances and improved accuracy through boosting techniques.

e) Voting Classifier: Combined multiple models to enhance overall reliability.

Model Evaluation

Model performance was measured using various evaluation metrics:

- a) Accuracy: Assessed the correctness of predictions.
- b) Precision and Recall: Focused on model performance for individual classes.
- c) F1-Score: Balanced precision and recall for a more comprehensive metric.
- d) AUC-ROC Curve: Evaluated the ability of the models to distinguish between classes.

K-fold cross-validation was used to improve reliability, and results were compared across the OASIS1 and OASIS3 datasets.

Explainability

To make model predictions more interpretable, explainability techniques were employed:

- a) Global Analysis: Showed how individual features contributed to overall predictions.
 - b) Local Analysis: Provided explanations for specific predictions, aiding in understanding patient-specific risk factors.
- SHapley Additive exPlanations (SHAP) were utilized to generate these insights.

Addressing Class Imbalance

Class imbalance, especially prominent in OASIS1, was mitigated using oversampling techniques. This ensured better representation of all dementia stages, resulting in improved model generalization.

Results

The findings of this study illustrate the capability of machine learning models to predict Alzheimer's Disease (AD) stages using features from the OASIS datasets. Key results are summarized below:

Model Performance on OASIS1 (Cross-Sectional Data)

- a) Random Forest:
 - Achieved an overall accuracy of 68%.
 - Precision and recall for the majority class (Class 0: Normal) were 0.78 and 0.80, respectively.
 - Struggled to predict minority classes (Class 1: Mild Dementia, Class 2: Severe Dementia) due to class imbalance.
- b) XGBoost:
 - Improved accuracy to 74%, with better handling of minority classes.
 - Precision and recall for Class 1 and Class 2 showed moderate improvement compared to Random Forest.
 - Demonstrated the effectiveness of boosting techniques for imbalanced datasets.
- c) AUC-ROC Analysis:
 - Class 0 (Normal): AUC = 0.87
 - Class 1 (Mild Dementia): AUC = 0.72
 - Class 2 (Severe Dementia): AUC = 0.92

These results highlight the strong performance of the model in detecting severe dementia while emphasizing the difficulty in accurately identifying mild dementia stages.

Model Performance on OASIS3 (Longitudinal Data)

- a) Random Forest:
 - Achieved an accuracy of 93%, benefiting from longitudinal tracking.
 - Precision and recall for the most common class (Class 0: Normal) were 0.97, indicating near-perfect classification.
 - Despite high overall accuracy, challenges in predicting minority classes persisted.
- b) XGBoost:
 - Maintained a high accuracy of 91%, with robust performance across dementia stages.

- Demonstrated better balance across classes compared to Random Forest.

c) AUC-ROC Analysis:

- Class 0 (Normal): AUC = 0.97

- Class 1 (Mild Dementia): AUC = 0.95

- Class 2 (Severe Dementia): AUC = 0.82

These results underscore the substantial impact of longitudinal data in enhancing model performance, particularly for Class 1 (Mild Dementia), compared to cross-sectional data.

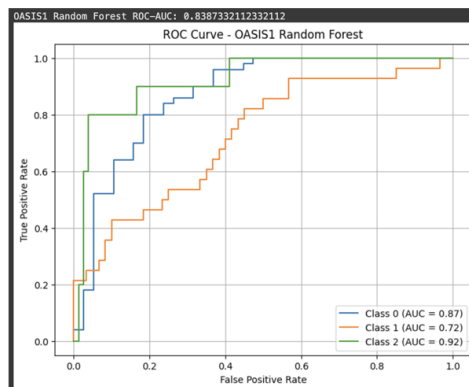


Figure 1 ROC OASIS1

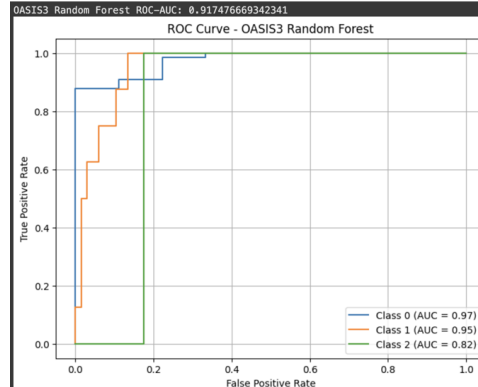


Figure 2 ROC OASIS3

Feature Importance

- MMSE emerged as the most predictive feature across both datasets, underlining its clinical significance.
- nWBV was strongly associated with cognitive decline and had high predictive power.
- Features such as age, eTIV, and ASF consistently contributed to model accuracy.
- Longitudinal variables in OASIS3, including MR delay and visit timing, were critical in improving model predictions.

Addressing Class Imbalance

- Random Oversampling enhanced performance for minority classes in OASIS1 but did not achieve complete balance.
- Boosting algorithms like XGBoost reduced performance gaps further, achieving moderate balance across classes.

Explainable AI Insights

d) SHapley Additive exPlanations (SHAP): provided detailed insights into model behavior:

- Global Analysis: MMSE and nWBV were consistently identified as top contributors to predictions.
 - Local Analysis: Individual predictions were explained to understand specific patient risks, enhancing clinical relevance.
- These results demonstrate the promise of machine learning models, particularly with longitudinal data, for improving early diagnosis of Alzheimer's Disease. Despite these advances, challenges like class imbalance and minority class prediction require further research.

Discussion

The findings of this study emphasize both the potential and the challenges associated with employing machine learning models for the early diagnosis of Alzheimer's Disease (AD). By leveraging the OASIS datasets, the study identified key predictors of dementia stages and demonstrated strong model performance, particularly with longitudinal data. Nonetheless, certain limitations and areas for improvement were evident.

Key Findings

a) Impact of Longitudinal Data:

The enhanced performance of models on the OASIS3 dataset highlights the critical role of longitudinal data in understanding disease progression. Features such as MRI delay and multiple visit data offered valuable insights into cognitive and

structural changes over time. This contributed to higher overall accuracy and improved classification of mild dementia stages.

Feature Importance:

MMSE and nWBV consistently emerged as the most significant predictors across datasets, reaffirming their clinical importance in assessing cognitive decline and brain atrophy. The inclusion of MRI-based features like eTIV and ASF further boosted model performance. Refer to the figure below.

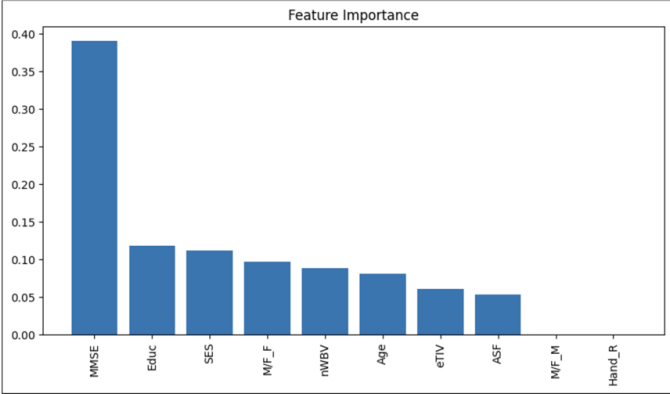


Figure 3 Feature Importance Bar Chart

Model Performance:

Random Forest and XGBoost effectively handled high-dimensional data, with XGBoost demonstrating superior capability in addressing class imbalance. Ensemble techniques like Voting Classifiers enhanced model robustness; however, individual models still faced difficulties in predicting minority classes.

OASIS1 XGBoost:

	precision	recall	f1-score	support
0	0.77	0.92	0.84	50
1	0.68	0.46	0.55	28
2	0.67	0.60	0.63	10
accuracy			0.74	88
macro avg	0.71	0.66	0.67	88
weighted avg	0.73	0.74	0.72	88

Confusion Matrix:

[[46	4	0]
[12	13	3]
[2	2	6]]

OASIS3 XGBoost:

	precision	recall	f1-score	support
0	0.94	0.97	0.96	66
1	0.57	0.50	0.53	8
2	0.00	0.00	0.00	1
accuracy			0.91	75
macro avg	0.50	0.49	0.50	75
weighted avg	0.89	0.91	0.90	75

Confusion Matrix:

[[64	2	0]
[4	4	0]
[0	1	0]]

Figure 4 XGBoost Results

OASIS1 Random Forest:

	precision	recall	f1-score	support
0	0.78	0.80	0.79	50
1	0.52	0.50	0.51	28
2	0.60	0.60	0.60	10
accuracy			0.68	88
macro avg	0.63	0.63	0.63	88
weighted avg	0.68	0.68	0.68	88

Confusion Matrix:

[[40	10	0]
[10	14	4]
[1	3	6]]

OASIS3 Random Forest:

	precision	recall	f1-score	support
0	0.97	0.97	0.97	66
1	0.67	0.75	0.71	8
2	0.00	0.00	0.00	1
accuracy			0.93	75
macro avg	0.55	0.57	0.56	75
weighted avg	0.92	0.93	0.93	75

Confusion Matrix:

[[64	2	0]
[2	6	0]
[0	1	0]]

Figure 5 Random Forest Results

Explainability:

The use of SHapley Additive exPlanations (SHAP) provided crucial insights into feature contributions, ensuring transparency and interpretability. This step is essential for fostering trust in machine learning tools among healthcare professionals.

Challenges

Class Imbalance:

Class imbalance posed a persistent challenge, especially for minority classes such as Mild Dementia and Severe Dementia. Despite preprocessing approaches like random oversampling, this issue impacted on the generalizability and reliability of the models.

a) Minority Class Prediction:

Boosting algorithms improved the balance between classes but could not entirely resolve the lower precision and recall for minority classes. This limitation reflects the inherent challenges of medical datasets, where certain conditions are underrepresented.

Generalizability:

The models' performance is primarily based on the OASIS datasets, and their applicability to external datasets remains uncertain. Testing on independent datasets is essential for broader validation and generalization.

Clinical Implications:

The results of this study have meaningful implications for clinical practice:

- a) Early Intervention: Improved classification of dementia stages, particularly mild dementia, could facilitate timely and targeted therapeutic interventions.
- b) Personalized Medicine: SHAP-based insights enable patient-specific risk factor identification, aiding in tailored treatment plans.
- c) Resource Optimization: Machine learning tools can assist clinicians in prioritizing cases for diagnostic evaluations, easing the strain on healthcare systems.

Future Directions

a) Advanced Techniques:

- Incorporating synthetic data generation methods like SMOTE to improve class balance.
- Exploring advanced deep learning architectures to capture complex patterns and relationships within the data.

b) External Validation:

- Testing the models on diverse, independent datasets to evaluate their robustness and generalizability.

c) Multi-Modal Data Integration:

- Integrating additional data sources, such as genetic markers and lifestyle factors, to enhance predictive accuracy.

d) Algorithmic Fairness:

- Assessing model performance across different demographic subgroups to ensure fairness and mitigate biases.

This study highlights the significant potential of machine learning models for enhancing the early diagnosis of Alzheimer's Disease. Addressing challenges like class imbalance and ensuring clinical adoption through interpretability, fairness, and validation are essential steps toward broader application.

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